

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Date: 01/08/2026

Experiment 8: Linear Regression (LR) Algorithm

Aim

To implement the Linear Regression (LR) algorithm in Python for predicting continuous values using an appropriate dataset.

Procedure

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Create the Linear Regression model.
5. Train the model using the training data.
6. Predict the target values for the test data.
7. Evaluate the model using Mean Squared Error (MSE) and R^2 score.
8. Display the prediction results.

Algorithm

1. Start.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Initialize the Linear Regression model.
5. Train the model using the training data.
6. Predict the output for the test data.
7. Calculate the Mean Squared Error (MSE) and R^2 score.
8. Display the predicted values and evaluation metrics.
9. Stop.

Code

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

data = load_diabetes()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Values:")
print(y_pred)

print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

print("R2 Score:", r2_score(y_test, y_pred))
```

Output

```
... Predicted Values:
[139.5475584 179.51720835 134.03875572 291.41702925 123.78965872
 92.1723465 258.23238899 181.33732057 90.22411311 108.63375858
 94.13865744 168.43486358 53.5047888 206.63081659 100.12925869
130.66657085 219.53071499 250.7803234 196.3688346 218.57511815
207.35050182 88.48340941 70.43285917 188.95914235 154.8868162
159.36170122 188.31263363 180.39094033 47.99046561 108.97453871
174.77897633 86.36406656 132.95761215 184.53819483 173.83220911
190.35858492 124.4156176 119.65110656 147.95168682 59.05405241
71.62331856 107.68284704 165.45365458 155.00975931 171.04799096
61.45761356 71.66672581 114.96732206 51.57975523 167.57599528
152.52291955 62.95568515 103.49741722 109.20751489 175.64118426
154.60296242 94.41704366 210.74209145 120.2566205 77.61585399
187.93203995 206.49337474 140.63167076 105.59678023 130.70432536
202.18534537 171.13039501 164.91423047 124.72472569 144.81030894
181.99635452 199.41369642 234.21436188 145.95665512 79.86703276
157.36941275 192.74412541 208.89814032 158.58722555 206.02195855
107.47971675 140.93598906 54.82129332 55.92573195 115.01180018
78.95584188 81.56087285 54.37997256 166.2543518 ]
Mean Squared Error: 2900.193628493482
R2 Score: 0.4526027629719195
```

Result

The Linear Regression (LR) algorithm was successfully implemented in Python. The model was trained and tested using the Diabetes dataset, and its performance was evaluated using Mean Squared Error (MSE) and R^2 score.